Vector Search and Hybrid Retrieval

This document explains how semantic (vector) search is combined with keyword search in our retrieval pipeline.

Embeddings are stored in Postgres using the pgvector extension. An ivfflat index approximates nearest-neighbor search over cosine distance, trading a small amount of recall for much faster queries at scale compared to an exact scan.

Pure vector search misses exact keyword matches, such as an error code or a function name a user pastes verbatim. To cover this, results from a Postgres full-text search (tsvector/GIN index) are combined with the vector results.

The two ranked lists are merged with Reciprocal Rank Fusion (RRF): each document's score is the sum of 1/(k + rank) across every list it appears in, with k=60. Documents that rank well in both searches float to the top even if neither list alone ranked them first.